

Justify your answers and show your work.

Evaluate each algebraic expression for the given value of the variable.

1. $3 - 4(5 + x)^2$, for $x = -6$

2. $\frac{2x+y}{xy-2x}$, for $x = -1$ and $y = 4$

Simplify the algebraic expression. Write all exponents as positive.

3. $8 - 7[-\frac{1}{2}(6 - 2) + 1]$

4. $\frac{5}{(7-12)^2} + \frac{1-3}{5}$

5. $\left(-\frac{2}{y}\right)^2$

6. $(9x^4)^{1/2}$

7. $(x^5x^2)^{-2}$

8. $(-5x^4y)(-6x^7y^{11})$

9. $\frac{8xy^4}{16x^2y^{-6}}$

10. $\sqrt{\frac{1}{49}}$

11. $5\sqrt{18} - 3\sqrt{2}$

Perform the indicated operation and simplify.

12. $(x^5 - 7x^2 + 5x) - (-3x^5 + x^2 + 13)$

13. $(4x^2y - 3xy) + (2x^2y - xy)$

14. $(5x - 2)(x^2 + x - 3)$

15. $(7x + 2y)^2$

Factor Completely.

16. $x^2 + 4x - 12$

17. $4x^3 - 8x^2$

18. $5x^3 + 20x^2 - 60x$

19. $9y^2 - 100$

Factor by grouping.

20. $x^3 + 6x^2 - 2x - 12.$